

Zhiyi Chen

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Klaus Advanced Computer Building, room 3337, Atlanta, GA, US

Education

Georgia Institute of Technology

Ph.D. Candidate in Computer Science; Advisor: Alberto Dainotti
Master of Science in Computer Science; Concentration: Machine Learning
GPA: 4.0/4.0

Atlanta, GA

Jan 2021 - present

Jan 2021 - May 2025

Tsinghua University

Bachelor of Science in Computer Science and Technology
GPA: 3.73/4.0

Beijing, China

Aug 2016 - Jul 2020

Selected Research Experience

Internet Measurement + LLM: Leveraging Web Presence and AI to Improve AS Classification

Georgia Tech

Research Assistant; Advisor: Alberto Dainotti, Cecilia Testart, Zachary S. Bischof

Nov 2024 - May 2025

- Developed an AS business classification dataset using LLMs to analyze websites, achieving an estimated 75.4% improvement over the state-of-the-art.

Internet Measurement: Improving the Inference of Sibling ASes

Georgia Tech

Research Assistant; Advisor: Alberto Dainotti, Cecilia Testart, Zachary S. Bischof

Aug 2021 - Dec 2022

- Identified two root causes that led to systematic inaccuracies in the state-of-the-art AS-to-organization methodology.
- Built an improved method on sibling relations, correcting mappings of 12.5% of organizations with sibling ASes.

Machine Learning: Automatic Tabular Data Preprocessing Pipeline

Georgia Tech

Research Assistant; Advisor: Xu Chu

Jan 2021 - Aug 2021

- Proposed a methodology to automatically and efficiently search for a data pre-processing pipeline for a given tabular dataset while maximizing the downstream machine learning model at the same time.
- Achieved the best test accuracy on 15 real-world datasets and improved the model's test accuracy by up to 6.6%.

Machine Learning: Continual Learning Algorithm Design

Peking University

Research Intern; Advisor: Tong Lin

Apr 2019 - Feb 2021

- Proposed a novel gradient descent training framework for mitigating catastrophic forgetting in continual learning.
- Improved the average test accuracy by up to 7.9% and reduced forgetting by up to 15.4% on four datasets.

Work Experience

Catchpoint Systems, Inc.

Remote

CTG Intern; Advisor: Gavin Lynch

Jun 2024 - Aug 2024

- Explored complicated Internet test data from test nodes all over the world towards cloud and content providers.
- Designed automatic approaches for detecting false positives in inferred cloud outages and developed methods for detecting cloud performance degradations.

Coalition, Inc.

Remote

Research Intern; Advisor: Daniel Woods, Morgan Hervé-Mignucci

Jun 2023 - Aug 2023

- Leveraged public Internet measurement platforms and GIS tools to infer and visualize geographic physical locations of datacenters that host policyholder domains.
- Explored the aggregation of cyber risks on physical infrastructure, especially critical points such as datacenters.

Publications

- Chen, Z., Bischof, Z. S., Testart, C., Dainotti, A. **Improving the Inference of Sibling Autonomous Systems**, published at the Passive and Active Measurement Conference (PAM) 2023.
 - Our dataset won *PAM 2023 Best Community Artifact Award*.
 - A lightning talk presented at NANOG (North American Network Operators' Group) 87 in Atlanta.
- Li, P., Chen, Z., Chu, X., Rong, K. **DiffPrep: Differentiable Data Preprocessing Pipeline Search for Learning over Tabular Data**, published at The ACM Special Interest Group on Management of Data (SIGMOD) 2023.
- Chen, Z., Lin, T. **Principal Gradient Direction and Confidence Reservoir Sampling for Continual Learning**, published at the International Conference on Artificial Neural Networks (ICANN) 2021.